



Comparative retrospective outcomes of endoscopic ultrasound-guided gastroenterostomy in patients with gastric outlet obstruction and post-surgical anatomy

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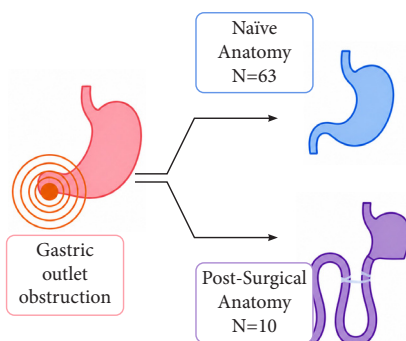
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Comparative retrospective outcomes of endoscopic ultrasound-guided gastroenterostomy in patients with gastric outlet obstruction and post-surgical anatomy

There is limited data evaluating outcomes of EUS-guided gastroenterostomy in patients with post-surgical anatomy

- Single tertiary center analysis
- Retrospective comparison with historical control group
- Excluded: adjunctive procedures, such as biliary drainage



	Cases (n=10)	Controls (n=63)	P-value
Clinical success, n (%)	90.0%	90.5%	1.000
Technical success, n (%)	90.0%	96.8%	0.362
Overall AE, n (%)	7.9%	10%	1.000
Procedural duration, min	61	45	0.037

Although increased complexity and limited sample size should be considered, our comparative data suggests that EUS-gastroenterostomy seems equally effective in patients with post-surgical anatomy.

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Background/Aims: Data on endoscopic ultrasound (EUS)-guided gastroenterostomy (EUS-GE) are mainly derived from patients with naïve anatomy. This study aimed to evaluate the outcomes of EUS-GE in patients with post-surgical anatomy and compare these findings with those from a cohort of patients with naïve anatomy.

Methods: A retrospective tertiary center analysis was performed of consecutive EUS-GE procedures. The wireless-EUS-GE-simplified technique (WEST) was employed for EUS-GE. Clinical success was defined as achieving a postprocedural gastric outlet obstruction scoring system score of ≥ 2 .

Results: Overall, 10 patients with post-surgical anatomy and 63 sequential historical controls with naïve anatomy were included from the same time frame (2019–2022). Surgery consisted predominantly of Whipple resection ($n=3$) or partial gastrectomy ($n=2$). Clinical success was achieved in 90.0% and 90.5% of patients in the post-surgical and naïve anatomy groups, respectively ($p=1.000$). Technical success (90.0% vs. 96.8%, $p=0.362$), median hospital stay (8 vs. 6 days, $p=0.397$), adverse event rate (10.0% vs. 7.9%, $p=1.000$), and median follow-up duration (186 vs. 136 days, $p=0.303$) were comparable between the two groups. However, the median procedure duration was significantly longer in the post-surgical anatomy group compared to that in the naïve group (61 vs. 45 minutes, $p=0.037$).

Conclusions: Although increased technical complexity and a limited sample size should be considered, our comparative data suggest that EUS-GE was equally effective in patients with post-surgical anatomy.

Keywords: Endoscopic ultrasound-guided gastrojejunostomy; Lumen-apposing metal stent; Postoperative anatomy; Surgically altered anatomy

INTRODUCTION

Gastric outlet obstruction (GOO) is a common manifestation of gastrointestinal malignancies, and its management has undergone significant advancements over the past decade. Among the numerous developments in endoscopy, endoscopic ultrasound (EUS)-guided gastroenterostomy (EUS-GE) has emerged as an established and valuable modality for the management of both malignant and benign GOO.¹⁻³ This approach is associated with a reduced need for re-intervention and superior clinical success rates compared to enteral stenting,⁴⁻⁶ while demonstrating equal efficacy, fewer adverse events (AEs), and a shorter median hospital stay compared to surgery.⁷⁻¹¹ Recently, prospective and randomized data from EUS-GE have become available, confirming these exact patterns.¹²⁻¹⁴ In addition to improved data quality, therapeutic EUS continues to advance with expanding indications, including the treatment of afferent loop syndrome, same-session “double EUS-bypass” procedures with simultaneous EUS-guided bile duct drainage, and the use of EUS-GE in non-conventional situations, such as post-surgical anatomy.¹⁵⁻¹⁹ Unfortunately, in many patients who have undergone previous gastrointestinal surgery, oncological disease recurrence may occur over time, especially after pancreatic cancer surgery, where the 5-year survival rate is only approximately 20%.²⁰ As local disease recurrence may interfere with adequate food intake and quality of life, patients with post-surgical anatomy may require endoscopic palliation of GOO-related symptoms. In this context, a minimally invasive approach seems optimal for prevent-

ing delays in systemic oncological treatment and optimizing the short-term quality of life. To date, limited data are available regarding the use of EUS-GE in patients with post-surgical anatomy, and the applicability of previous outcome data on EUS-GE to this population remains unclear.

This study aimed to compare the efficacy and safety of EUS-GE in patients with naïve versus post-surgical anatomy.

METHODS

We performed a single-center retrospective study that included all EUS-GE procedures for post-surgical GOO from an institutional database. The inclusion criteria were as follows: (1) malignant or benign GOO following upper gastrointestinal surgery, including Whipple resections and Roux-en-Y and/or Billroth I and II reconstructions; (2) endoscopic or radiological confirmation of benign or malignant gastroduodenal and biliary stenosis; and (3) treatment with EUS-GE. No additional techniques (such as EUS-guided biliary drainage) were employed to minimize potential confounding from adjunctive procedures. These cases were compared with a historical cohort of GOO in patients with naïve anatomy from the same period, which served as the control group.⁶ For both approaches, data were collected on age, sex, American Society of Anesthesiologists (ASA) score, underlying disease, disease stage, and presence of ascites and peritoneal carcinomatosis.

Endpoints

The primary endpoint was clinical success, defined as a GOO score system (GOOSS) of ≥ 2 ^{6,9,11,21,22} or the ability to tolerate at least a soft diet without vomiting. Food intake grade was obtained from electronic patient files and/or during subsequent follow-ups. Technical and clinical success, as well as food intake grading, are described in greater detail under the “Efficacy” section. The secondary outcome was safety, with AE stratified as mild, moderate, severe, or fatal according to the American Society for Gastrointestinal Endoscopy lexicon for AEs.²³ Lumen-apposing metal stent (LAMS) maldeployment was classified according to the system proposed by Ghandour et al.²⁴ The overall anesthesiology risk was assessed using the ASA score. Hospital stay, time to oral intake, and time to full diet (GOOSS of 4) were also recorded.

Follow-up

Gastroenterostomy dysfunction was defined as recurrence of obstructive symptoms (GOOSS ≤ 1) after former clinical success, with confirmation of recurrent GOO by endoscopy or imaging studies. Mechanical obstructions during follow-up located downstream of the gastroenterostomy site without evidence of EUS-GE dysfunction were recorded as “distal obstructions.”

Procedure: EUS-GE

All procedures were performed under general anesthesia, with prophylactic administration of broad-spectrum antibiotics. For each EUS-GE procedure, the wireless EUS-GE simplified technique (WEST) was used.^{6,11,25,26} A standard gastroscope was introduced, followed by guidewire advancement through the stricture or anastomosis into the small bowel or alimentary limb (Fig. 1). A 7-French oro-enteric catheter was then advanced over a guidewire, after which the gastroscope was withdrawn. A linear echo-endoscope was then carefully introduced, and varying amounts of saline were infused using 50 cc syringes (with or without blue dye and antispasmodics). After adequate small bowel dilation and optimization of the scope position, a 15×10 mm or 20×10 mm LAMS was deployed into the target segment of the small bowel using freehand insertion and pure cutting current (Autocut, effect 5; 100–120 W). Following deployment of the LAMS, adequate positioning was ascertained using fluoroscopy, EUS, visual confirmation, or a combination of approaches.

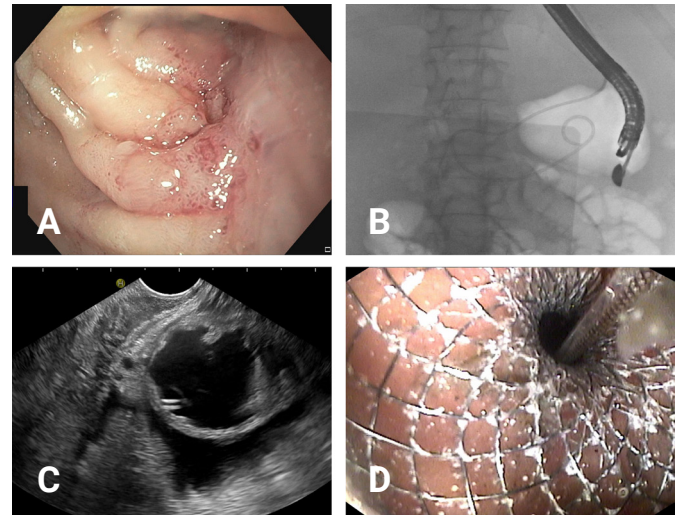


Fig. 1. (A) Endoscopic confirmation of a malignant stenosis at the surgical anastomosis. (B) Fluoroscopic image displaying the direction of the linear endoscopic ultrasound (EUS) scope towards the oroenteric catheter. (C) EUS image confirming adequate small bowel dilation and an intraluminal oroenteric catheter. (D) Endoscopic image demonstrating successful deployment of the proximal flange and backflow of non-dyed saline.

Statistical analysis

Categorical variables were reported as frequencies (%), and Fisher’s exact test or chi-squared test was utilized to compare these variables. Continuous variables were reported as medians and interquartile range (IQR) or means±standard deviation. Student *t*-test and Mann-Whitney *U*-test were employed to compare normally and non-normally distributed continuous variables. Statistical significance was set at $p < 0.05$. IBM SPSS ver. 26.0 (IBM Corp.) and Excel ver. 2023 (Microsoft) were employed for statistical analyses.

Ethical considerations

Each patient provided written informed consent for the individual technical procedures. Institutional Review Board approval was obtained from the University Hospital Leuven (identifiers: s64254 and S68390).

RESULTS

Overall, 10 patients with a history of post-surgical anatomy and 63 sequential historical controls with naïve anatomy were included from the same study period (2019–2022). The surgical histories comprised Whipple resection ($n=3$), partial gastrecto-

my ($n=2$), bile duct resection with Roux-en-Y reconstruction ($n=2$), Billroth I ($n=1$), entero-enterostomy and re-anastomosis ($n=1$), and Roux-en-Y gastric bypass ($n=1$).

The mean age and sex distribution were similar, as were the prevalences of ascites and peritoneal carcinomatosis (Table 1). Pancreatic cancer was the most prevalent primary disease (47.6% vs. 10.0%) in the naïve anatomy group, whereas benign strictures were the most frequent underlying cause of GOO in

the post-surgical anatomy group (12.7% vs. 50.0%).

The ASA score distribution was similar in both groups (ASA-I, 1.6% vs. 0.0%; ASA-II, 27.0% vs. 40.0%; ASA-III, 42.9% vs. 40.0%; and ASA-IV, 28.6% vs. 20.0%).

Efficacy

Clinical success, analyzed on an intention-to-treat basis, was achieved in 90.0% of patients with post-surgical and 90.5% of

Table 1. Baseline characteristics

Variable	Overall cohort ($n=73$)	Naïve anatomy ($n=63$)	Post-surgical anatomy ($n=10$)
Age (yr)	66±11.9	65.8±12.0	68±11.5
Sex			
Female	23 (31.5)	20 (31.7)	3 (30.0)
ASA score			
I	1 (1.4)	1 (1.6)	0 (0)
II	21 (28.8)	17 (27.0)	4 (40.0)
III	31 (42.5)	27 (42.9)	4 (40.0)
IV	20 (27.4)	18 (28.6)	2 (20.0)
Median follow-up (day)	136 (61–233)	136 (61–233)	186 (81–672)
Primary disease			
Pancreatic cancer	31 (42.5)	30 (47.6)	1 (10.0)
Benign disease	13 (17.8)	8 (12.7)	5 (50.0)
Pancreatitis-related	8 (11.0)	7 (11.1)	1 (10.0)
Anastomotic stricture	2 (2.7)	0 (0)	2 (20.0)
Superior mesenteric artery syndrome	1 (1.4)	1 (1.6)	0 (0)
Peptic stricture	1 (1.4)	0 (0)	1 (10.0)
Postoperative gastroparesis	1 (1.4)	0 (0)	1 (10.0)
Gastric cancer	7 (9.6)	5 (7.9)	2 (20.0)
Cholangiocarcinoma	5 (6.8)	4 (6.3)	1 (10.0)
Duodenal cancer	5 (6.8)	5 (7.9)	0 (0)
Non-small cell lung cancer	4 (5.5)	4 (6.3)	0 (0)
Colorectal cancer	3 (4.1)	3 (4.8)	0 (0)
Melanoma	1 (1.4)	0 (0)	1 (10.0)
Oesophageal cancer	1 (1.4)	1 (1.6)	0 (0)
Cervical cancer	1 (1.4)	1 (1.6)	0 (0)
Neuroendocrine tumor	1 (1.4)	1 (1.6)	0 (0)
Transitional cell carcinoma	1 (1.4)	1 (1.6)	0 (0)
Disease stage			
Local invasion	23 (31.5)	21 (33.3)	2 (20.0)
Local and peritoneal metastases	10 (13.7)	10 (15.9)	0 (0)
Local and liver metastases	13 (17.8)	13 (20.6)	0 (0)
Diffuse metastatic	14 (19.2)	11 (17.5)	3 (30.0)
Disease manifestations			
Ascites	17 (23.3)	15 (23.8)	2 (20.0)
Peritoneal carcinomatosis	24 (32.8)	21 (33.3)	3 (30.0)

Values are presented as mean±standard deviation, number (%), or median (inclusive interquartile range). ASA, American Society of Anaesthesiologists.

those with naïve anatomy ($p=1.000$). Technical success rates were similar in both groups (90.0% and 96.8%, respectively). No differences in median time to clinical success (1 [IQR, 1–2] vs. 1 [IQR, 1–1], $p=0.334$) or median time to oral intake (1 [IQR, 1–1] vs. 1 [IQR, 0–1], $p=0.430$) were observed between patients with post-surgical and naïve anatomy. Furthermore, the median hospital stay was comparable between the two groups (8 [IQR, 2–16] vs. 6 [IQR, 2.8–11], $p=0.397$) (Table 2). The median procedural duration was significantly longer in the post-surgical anatomy group (61 [54–86] vs. 45 [30–65] minutes, $p=0.037$).

Safety

Overall, AEs were observed in 10.0% and 7.9% of patients in the post-surgical and naïve anatomy groups, respectively (10.0% [$n=1$] vs. 7.9% [$n=5$], $p=1.000$). One LAMS maldeployment (grade: severe, maldeployment type IV) occurred in the post-surgical anatomy group, for which surgical revision was performed at the patient's and referring physician's request. In the naïve anatomy group, mild and moderate AEs occurred in one (1.6%) and three patients (4.8%), respectively. One severe AE occurred (1.6%), consisting of a late LAMS dislocation more than two hours after the procedure, which required surgical revision due to the postprocedural time interval. Mild AE

consisted of postprocedural pain ($n=1$), while moderate AE comprised sepsis ($n=1$), successfully managed with broad-spectrum antibiotics, and self-limiting bleeding at the anastomosis site ($n=2$).

Follow-up

During a median follow-up of 136 (IQR, 61–233) days, two dysfunctions were identified in the naïve anatomy group (3.2%), attributed to ingrowth ($n=1$) or delayed LAMS migration ($n=1$). The median follow-up in the post-surgical anatomy group was 186 days (IQR, 81–672; $p=0.303$). No dysfunction was observed ($p=1.000$), and six patients died from malignant disease within 90 days. One patient with benign GOO in the postsurgical group died from an unrelated cancer. Among three patients with benign disease in the post-surgical anatomy group, yearly endoscopic or radiological follow-ups for LAMS patency were performed, resulting in one elective LAMS exchange due to ingrowth. One incidental nondisease-related death occurred in the same group.

DISCUSSION

In this single-center retrospective analysis, EUS-GE in patients

Table 2. Outcomes

	Naïve anatomy ($n=63$)	Post-surgical anatomy ($n=10$)	OR (95% CI)	p -value
Efficacy				
Primary outcomes				
Technical success	61 (96.8)	9 (90.0)	3.4 (0.28–41)	0.362
Clinical success	57 (90.5)	9 (90.0)	1.1 (0.11–9.8)	1.000
Median time to clinical success (day)	1 (1–1)	1 (1–2)		0.334
Median time to oral intake (day)	1 (0–1)	1 (1–1)		0.430
Median procedure duration (min)	45 (30–65)	61 (54–86)		0.037
Dysfunction (after initial success)	2 (3.2)	0 (0)	1.2 (0.052–26)	1.000
Median time to dysfunction (day)	286 (60.8–295) ^{a)}	NA ^{a)}		NA ^{a)}
Secondary outcomes				
Median hospital stay (day)	6 (2.8–11)	8 (2.0–16)		0.397
Safety				
Overall AEs	5 (7.9)	1 (10.0)	1.3 (0.14–12)	1.000
ASGE AE severity grading system				
Mild	1 (1.6)	0 (0)		1.000
Moderate	3 (4.8)	0 (0)		1.000
Severe	1 (1.6)	1 (10.0)		1.000
Fatal	0 (0)	0 (0)		1.000

Values are presented as number (%) or median (interquartile range).

OR, odds ratio; CI, confidence interval; ASGE, American Society for Gastrointestinal Endoscopy; AE, adverse event.

^{a)}Observed in the naïve anatomy group only.

with post-surgical anatomy yielded outcomes comparable to those observed in patients with naïve anatomy. Although increased anatomical complexity necessitates adaptation of existing techniques to specific anatomical scenarios, our findings indicate that EUS-GE offers effective symptom palliation in this population.

Recent evidence, including a video report, demonstrates that EUS-GE can be safely and effectively performed in patients following duodenectomy. In this patient, direct EUS-GE was performed using 19-G needle-guided saline instillation.²⁷ In our study, small bowel loop distension was achieved by saline insertion via an orojejunal catheter, followed by freehand placement according to the WEST approach.²⁵ Another area of debate concerns the use of balloon dilation following LAMS placement, which varies widely across different regions.²⁸ Balloon dilatation is not routinely performed at our center due to the absence of a clear additional benefit.^{6,11,26} Furthermore, the accompanying fluoroscopy images from these published cases support our observation that catheter placement and LAMS deployment in post-surgical anatomy differ from procedures in patients with naïve anatomy. The direction of stent insertion, target bowel location, and ease of small bowel target acquisition may vary greatly, depending on the exact underlying configuration and small bowel distensibility. This increased complexity was also reflected in the significantly prolonged duration of the procedure (61 vs. 45 minutes, $p=0.037$). However, once an adequately dilated small bowel loop is obtained, the insertion technique and postprocedural care are identical to those of EUS-GE in naïve anatomy.

A recent large multicenter analysis confirmed that LAMS use in post-surgical anatomy is effective regardless of the indication, albeit with a non-negligible risk of AEs (12%), most of which were classified as AGREE grade IIIa or IIIb.²⁹

Enteral stenting has been proposed as a safer alternative; however, recent comparative data demonstrated similar complication rates.^{6,14} Furthermore, enteral stenting for post-surgical GOO³⁰ appears suboptimal in the current EUS-GE era due to the high risk of clinical failure, inferior long-term outcomes, and an increased likelihood of reinterventions.^{6,13,14,30} Therefore, the selective use of enteral stenting seems appropriate, for instance, in patients with uncontrolled ascites or failed EUS-GE.³¹

Surgical alternatives are less desirable in the context of malignancy due to increased invasiveness and inferior outcomes regarding oral intake.^{11,32} The primary indication for surgical gastroenterostomy remains in patients with benign GOO,

particularly those with post-surgical anatomy and preserved functional status. One of the most commonly cited arguments in favor of surgery is the potentially superior long-term outcomes following (redo) surgical intervention. However, in our current series, five patients with post-surgical anatomy and benign GOO were treated with favorable long-term results, with no LAMS dysfunction observed over time. Three patients died due to unrelated causes, while the remaining two continued annual LAMS surveillance for more than 3 years. Similar long-term efficacy was observed in one of the first studies to evaluate EUS-GE in benign disease.³³ Twenty-two patients were retrospectively evaluated, with four (18.2%) undergoing EUS-GE due to a benign anastomotic stricture following either Billroth I–II surgery or Roux-en-Y gastrojejunostomy. After a median follow-up of 465.5 days, LAMS could be electively removed in 83.3% of the patients following the resolution of GOO, with a low (9%) overall need for repeat surgery. These data in benign disease indicate that EUS-GE can also provide durable long-term patency, with the added advantage of being fully reversible, making it an attractive treatment option for patients at high surgical risk, especially when GOO is expected to be temporary.^{33–35}

Limitations and strengths

Limitations of the current study include its retrospective and single-center design, as well as differences in underlying disease etiology between the two groups. Finally, the sample size was small ($n=10$). However, no larger datasets focusing particularly on the outcomes of EUS-GE in post-surgical anatomy are currently available. Considering the exact underlying surgical anatomy, we included patients who had undergone bile duct resection and Roux-en-Y reconstruction. Although one could argue that the gastroduodenal anatomy in these cases resembles that of patients with naïve anatomy, the possibility that extensive underlying enteric surgery may influence EUS-GE creation and subsequent outcomes cannot be excluded. Another strength of our current series is the inclusion of a comparator group, which enabled a more meaningful assessment of comparative outcomes. The extrapolation of these outcomes to low-volume settings is challenging, as these procedures are only performed by therapeutic EUS experts in high-volume centers.

In conclusion, our comparative data suggest that EUS-GE seems similarly effective for managing GOO in patients with either naïve or post-surgical anatomy. Although increased technical complexity should be considered, the outcomes underscore

the significant role of EUS-GE in managing GOO, irrespective of the anatomical status.

Conflicts of Interest

MB received trial support from Boston Scientific and Ovesco and held consultancy agreements with Dekra, Ovesco, and Prion Medical/Taewoong. GV received lecture fees from Boston Scientific and travel grants from Pentax Medical and Euromedical. RvW holds consultancy agreements with Boston Scientific and Cook Medical and receives speaker fees from Olympus. SVD holds the Cook and Boston Scientific Chair in Interventional Endoscopy and consultancy agreements with Cook, Pentax, and Olympus.

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Author Contributions

Conceptualization: MB; Data curation: MB, JC, SVD; Formal analysis: all authors; Methodology: MB, JC, SVD; Project administration: MB, JC; Supervision: GV, JS, RW, SVD; Visualization: MB; Writing—original draft: MB; Writing—review & editing: all authors.

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